

Module 6: Stable matches, *markets without prices or goods*

ECO364-THINKING STRATEGICALLY

Markets without goods or prices?

▶ Two-sided markets

- ▶ Employers-employees
- ▶ Hospitals-residents
- ▶ Universities-applicants
- ▶ Matchmaking

▶ Single-sided markets:

- ▶ Teaming up (roommates)
- ▶ Organ exchange

- ▶ Most of the peculiar features of these markets is that they don't involve "economic goods" in the traditional sense. Furthermore, for many of these markets, whatever that is traded or exchanged can't be put a price on.
- ▶ In fact, most of the allocation problems in these markets stem from precisely this fact (think organ about organ donations/exchanges).
- ▶ As economists we love prices because they are an efficient way of allocating goods, equating supply and demand and maximizing welfare.
- ▶ Auctions, the topic from the last module, are an example of this.
- ▶ So how do markets that don't have prices or standard economic goods work?
- ▶ In the case no actual market (institution) exist to carry out the trades (like organ exchange), how could we design such a market?

The market of hospitals and medical interns in the USA

► Phase I: The tragedy of the doctors.

- Our story begins at the end of the XIX century, with the introduction of residency requirements for recent medical school graduates.
- Fresh medical school graduates needed to find a hospital in which to pursue their internships.
- Residents played an important role in the staff of hospitals, so hospitals began competing with each other for the best graduates.
- To get a jump on the competition, hospitals kept moving up the dates on which they granted residency positions.
- This was regarded as costly and inefficient both by the hospitals, who had to appoint interns without knowing their final grades or class standings, and by the students and medical schools, who found that much of the senior year was disrupted by the process of seeking desirable appointments.
- By 1944, medical students in their third year of medical school (out of four) were interviewed for residency positions!
- The problem was solved in 1945 when the medical schools in the Association of American Medical Colleges (AAMC) undertook not to give out before some agreed date any information to hospitals regarding students' abilities.

The market of hospitals and medical interns in the USA

► Phase II: The tragedy becomes a farce

- A new problem appeared, and manifested itself in the waiting-period between the time offers of internships were first made, and the time students were required to accept them.
- Basically, the problem was that a student would be inclined to wait as long as possible before accepting the position he had been offered, in the hope of eventually being offered a preferable position.
- Students who were pressured into accepting offers were unhappy if they were ultimately offered a preferable position.
- Hospitals whose candidates waited until the last minute to reject them were unhappy if their preferred alternate candidates had in the meantime already accepted positions.
- Hospitals were unhappier still when a candidate who had indicated acceptance subsequently failed to fulfill his commitment after receiving a preferable offer.
- For 1945, it was resolved that hospitals should allow students ten days after an offer had been made to consider whether to accept or reject it.
- For 1946, it was resolved that there should be a uniform appointment date (July 1) on which offers should be tendered and that acceptance or rejection should not be required before July 8.
- By 1949, the AAMC proposed that appointments should be made by telegram at 12:01 AM (on November 15), with applicants not being required to accept or reject them until 12:00 Noon the same day.
- In 1950, the resolution again included a twelve-hour period for consideration, with the specific injunction that "Hospitals and/or students shall not follow telegrams of offers of appointment with telephone calls" until after the twelve-hour grace period.

The market of hospitals and medical interns in the USA

► Phase III: Arranged matching

- In order to avoid these problems and the costs they imposed, it was proposed in 1951, and ultimately agreed, that a more centralized matching procedure should be tried.
- Students would then rank in order of preference the hospital programs to which they had applied, hospitals would similarly rank their applicants, and all parties would submit these rankings to a central bureau, which would use this information to arrange a matching of students to hospitals, and inform the parties of the result.
- This procedure was used between 1953 and 1998, matching roughly 20,000 physicians a year. It was replaced by a new one in 1998.
- (The National Resident Matching Program (NRMP))
- Participation in this central mechanism, on the part of both hospitals and students, is voluntary.
- The success of a “good matching” between residents and hospitals is a significant factor in determining the extent of participation in the matching process and its success.
- The beauty of the procedure used by the American national resident matching program is that it leads to matches in which no pair is dissatisfied.
- After all, is not easy to arrive at matches that please everyone!

The market of hospitals and medical interns in the USA

- ▶ Phase III: Arranged matching
- ▶ For example:
- ▶ If Mark prefers Massachusetts General Hospital to Johns Hopkins Hospital, and Massachusetts General Hospital prefers Mark to Andrew, it would be unwise to send Mark to Johns Hopkins and Andrew to Massachusetts General.
- ▶ Such a matching will generate dissatisfaction for both Mark and Massachusetts General, and both of them will then have an incentive to [disregard the matching program](#).
- ▶ If too many cases like this multiply, then the entire system collapses!
- ▶ In contrast, in the NRMP, if Mark is matched with Johns Hopkins Hospital, then Massachusetts General Hospital must have been matched with residents whom it preferred to Mark, and mark has no justified complaint, because his preferred hospital simply preferred others to him; no injustice, unfairness or inefficiency is involved in the process.

Two-sided Problems

- ▶ The problem of matching elements from two different populations is not limited to the example of hospitals and residents.
 - ▶ Employers-employees
 - ▶ Universities-applicants
 - ▶ Matchmaking
- ▶ A matching problem between two different populations raises many questions. For example, for the hospital-resident problem:
 - ▶ What is the best definition of a “stable placement” of candidates to hospital positions?
 - ▶ Do such an stable placement exist?
 - ▶ How to find it?
 - ▶ Can a hospital or candidate obtain a more satisfactory placement by lying about their preferences?
- ▶ In 1962, David Gale and Lloyd Shapley published a paper in which they defined the matching problem, the concept of a “stable matching” and they proved that stable matchings always exist and gave a procedure or algorithm to find them.
- ▶ Several years later, Alvin Roth connected the Gale-Shapley algorithm result to the algorithm used to match residents and hospitals in the US.
- ▶ Shapley and Roth were awarded the 2012 Nobel Prize for these contributions.
- ▶ We will begin analyzing the matching problem in the original setting of Gale and Shapley’s 1962 paper, a setting which is also a fun one: [the marriage problem!](#)

Welcome to the Spin-Dating Club!



- ▶ Ann, Betty, Carol, Daisy, Adam, Barry, Claud and Dave are participating in a Spin-Dating Club, each one of them looking to find a match with a member of the opposite sex.
- ▶ After taking some time to know each other, their preferences for the opposite sex can be described by the following table.

	Ann	Betty	Carol	Daisy
Adam	3	1	2	4
Barry	2	4	4	3
Claud	2	1	3	4
Dave	1	4	3	2

- ▶ The numbers represent how each individual ranks each member of the opposite sex, 1 being the highest and 4 the lowest.
- ▶ For example Betty ranks Adam first, Dave second, Claud third and Barry fourth.
- ▶ Similarly, Barry ranks Betty first, Ann second, Daisy third and Carol fourth.
- ▶ Notice that the preferences are **strict and complete**, in the sense that everybody is able to produce a full ranking of all members of the opposite sex.

Let's try our hand at matchmaking...

Adam - Ann, Barry - Betty, Claud - Carol, Dave - Daisy

- ▶ Adam would gladly swap Ann (3) for Daisy (1).
- ▶ But Daisy won't leave Dave (2) for Adam (4).
- ▶ Carol, too, would swap Claud (2) for Barry (1).
- ▶ But Barry won't leave Betty (1) for Carol (4).
- ▶ But Betty and Claud - that's another story.
- ▶ Claud prefers Betty (1) to Carol (3).
- ▶ And Betty would willingly switch Barry (4) for Claud (3).

	Ann	Betty	Carol	Daisy
Adam	3 1	4 1	2 3	1 4
Barry	2 4	1 4	4 1	3 3
Claud	2 2	1 3	3 2	4 1
Dave	1 3	4 2	3 4	2 2

Betty and Claud object to this matching!

Stability of a Matching

- ▶ A boy and a girl **object** to a match, if they prefer each other to their matched mates.
- ▶ A match is **stable** if there is no boy-girl pair that objects to it.
- ▶ Firstly, if there is such a match, how to find it?
- ▶ Secondly, we need to make sure that if we can find one in the previous example, it was not just because that example had “something” special, that is, we should be able to find a stable matching for every matching problem (every instance of boy-girl preferences).
- ▶ Can we do this?
- ▶ The answer is a resounding YES!
- ▶ The **deferred acceptance algorithm (DAA)** described by Gale and Shapley (1962) yields a stable matching.

Stability of a Matching

- ▶ Before elaborating on this algorithm, it should be noted that only two-sided markets are guaranteed to have a stable matching. One-sided markets are not guaranteed to have stable matchings.
- ▶ Example: We want to make **two two-person teams** from this group:

Al Bert Clive Gus

Their preferences

	Al	Bert	Clive	Gus
Al	*	1	2	3
Bert	2	*	1	3
Clive	1	2	*	3

- ▶ Suppose that Gus is matched with X.
- ▶ Clearly, X prefers any other partner to Gus.
- ▶ Another person, Y, has X on the top of his list, but is not matched with him.
- ▶ **The pair X and Y object to the matching.**
- ▶ This small proof means that for any pair of two-person teams we propose, the member of the team that is paired with Gus always prefers to be with any other teammate and in the other team there is always one person that prefers the partner of Gus. Thus these two guys **object the matching, and we can always find such a pair in this problem.**

Back to matchmaking: DAA-boys proposing

- Day one: every boy proposes to the first girl on his list.

	Ann	Betty	Carol	Daisy
Adam	3 1	4 1	2 3	1 4
Barry	2 4	1 4	4 1	3 3
Claud	2 2	1 3	3 2	4 1
Dave	1 3	4 2	3 4	2 2

Back to matchmaking: DAA-boys proposing

- ▶ Day one: every boy proposes to the first girl on his list.
- ▶ Every girl asks her favorite boy among those who proposed to come again the next day, and rejects all others.

	Ann	Betty	Carol	Daisy
Adam	3 1	4 1	2 3	1 4
Barry	2 4	1 X 4	4 1	3 3
Claud	2 2	1 3	3 2	4 1
Dave	1 3	4 2	3 4	2 2

Back to matchmaking: DAA-boys proposing

- ▶ Day one: every boy proposes to the first girl on his list.
 - ▶ Every girl asks her favorite boy among those who proposed to come again the next day, and rejects all others.
- ▶ Day two: all boys who were asked to come again do so, and rejected boys go to the next girl on their list.

	Ann	Betty	Carol	Daisy
Adam	3 1	4 1	2 3	1 4
Barry	2 4	X 1	4 1	3 3
Claud	2 2	1 3	3 2	4 1
Dave	1 3	4 2	3 4	2 2

Back to matchmaking: DAA-boys proposing

- ▶ Day one: every boy proposes to the first girl on his list.
 - ▶ Every girl asks her favorite boy among those who proposed to come again the next day, and rejects all others.
- ▶ Day two: all boys who were asked to come again do so, and rejected boys go to the next girl on their list.
 - ▶ Every girl asks her favorite boy among those who proposed to come again the next day, and rejects all others.

	Ann	Betty	Carol	Daisy
Adam	3 1	4 1	2 3	1 4
Barry	2 4 X	1 4	4 1	3 3
Claud	2 2	1 3	3 2	4 1
Dave	1 3	4 2	3 4	2 2

Back to matchmaking: DAA-boys proposing

- ▶ Day one: every boy proposes to the first girl on his list.
 - ▶ Every girl asks her favorite boy among those who proposed to come again the next day, and rejects all others.
- ▶ Day two: all boys who were asked to come again do so, and rejected boys go to the next girl on their list.
 - ▶ Every girl asks her favorite boy among those who proposed to come again the next day, and rejects all others.
- ▶ Day three: ditto.

	Ann	Betty	Carol	Daisy
Adam	3 1	4 1	2 3	1 4
Barry	2 4	X 4	4 1	3 3
Claud	2 2	1 3	3 2	4 1
Dave	1 3	4 2	3 4	2 2

Back to matchmaking: DAA-boys proposing

- ▶ Day one: every boy proposes to the first girl on his list.
 - ▶ Every girl asks her favorite boy among those who proposed to come again the next day, and rejects all others.
- ▶ Day two: all boys who were asked to come again do so, and rejected boys go to the next girl on their list.
 - ▶ Every girl asks her favorite boy among those who proposed to come again the next day, and rejects all others.
- ▶ Day three: ditto.
- ▶ Day four: ditto

	Ann	Betty	Carol	Daisy
Adam	3 1	4 1	2 3	1 4 X
Barry	2 4	1 4	4 1	3 3
Claud	2 2	1 3	3 2	4 1
Dave	1 3	4 2	3 4	2 2

Back to matchmaking: DAA-boys proposing

- ▶ Day one: every boy proposes to the first girl on his list.
 - ▶ Every girl asks her favorite boy among those who proposed to come again the next day, and rejects all others.
- ▶ Day two: all boys who were asked to come again do so, and rejected boys go to the next girl on their list.
 - ▶ Every girl asks her favorite boy among those who proposed to come again the next day, and rejects all others.
- ▶ Day three: ditto.
- ▶ Day four: ditto
- ▶ Day five: ditto

	Ann	Betty	Carol	Daisy
Adam	3 1	4 1	2 3	1 4 X
Barry	2 4	1 4	4 1	3 3
Claud	2 2	1 3	3 2	4 1
Dave	1 3	4 2	3 4	2 2

Back to matchmaking: DAA-boys proposing

- ▶ Day one: every boy proposes to the first girl on his list.
 - ▶ Every girl asks her favorite boy among those who proposed to come again the next day, and rejects all others.
- ▶ Day two: all boys who were asked to come again do so, and rejected boys go to the next girl on their list.
 - ▶ Every girl asks her favorite boy among those who proposed to come again the next day, and rejects all others.
- ▶ Day three: ditto.
- ▶ Day four: ditto
- ▶ Day five: ditto
- ▶ This is a stable matching: Adam-Carol, Barry-Daisy, Claud-Betty and Dave-Ann

	Ann	Betty	Carol	Daisy
Adam	3 1	4 1	2 3	1 4
Barry	2 4	1 4	4 1	3 3
Claud	2 2	1 3	3 2	4 1
Dave	1 3	4 2	3 4	2 2

Summary: DAA with boys proposing

- ▶ Every day, each boy proposes to his favorite girl among those who have not yet rejected him.
- ▶ Every girl rejects all her suitors, except the one she likes best.
- ▶ The algorithm ends when a girl has at most one suitor.
- ▶ For any boy-girl preference, the boy-proposing process ends with a stable matching.

DAA with girls proposing

► Day one

	Ann	Betty	Carol	Daisy
Adam	3 1	4 1	2 3	1 4
Barry	2 4	1 4	4 1	3 3
Claud	2 2	1 3	3 2	4 1
Dave	1 3	4 2	3 4	2 2

DAA with girls proposing

- ▶ Day one
- ▶ Day two
- ▶ The matching that we obtain with the girls proposing DAA is:
- ▶ Adam-Ann, Barry-Carol, Claud-Daisy, Dave-Berry

	Ann	Betty	Carol	Daisy
Adam	3 ¹	4 ¹	2 ³	1 ⁴
Barry	2 ⁴	1 ⁴	4 ¹	3 ³
Claud	2 ²	1 ³	3 ²	4 ¹
Dave	1 ³	4 ²	3 ⁴	2 ²

Comparison: Boys propose DAA vs Girls propose DAA

■ - girls propose

■ - boys propose

- ▶ A girl is feasible for a boy if there is a stable matching in which they are paired.
- ▶ A girl who rejected a boy during DAA with boy proposing is not feasible for him.
- ▶ Hence, The girl matches to a boy by DAA with boy proposing is his favorite among those feasible for him.
- ▶ Thus, the boy matched with a girl by DAA with boys proposing is the boy she least favors of those feasible for her.
- ▶ Boys prefer the stable matching obtained by the DAA with boys proposing and Girls prefer the stable matching obtained by the DAA with girls proposing.
- ▶ Good for the wooer and woe to the wooed!

	Ann	Betty	Carol	Daisy
Adam	3 1	4 1	2 3	1 4
Barry	2 4	1 4	4 1	3 3
Claud	2 2	1 3	3 2	4 1
Dave	1 3	4 2	3 4	2 2

What if the Boys propose DAA and Girls propose DAA overlap?

- ▶ Boys proposing and girls proposing yield the same stable matching.



- ▶ There is a single stable matching

Lonely Hearts Club

- ▶ Suppose that the number of boys is n , the number of girls is m , and $n > m$.
- ▶ In any stable matching, $n-m$ boys will remain unmatched.
- ▶ Does the fate of the unmatched boys depend on the matching? Might an unmatched boy be paired in a different stable matching?
- ▶ Alas for the bachelors, the answer is:

No!

- ▶ The unmatched boys are the same in every stable matching.

Lonely Hearts Club

- ▶ To show this we take a boy who is unmatched in BP-DAA.
- ▶ We've seen that a girl who rejects a boy in this process isn't paired with him in any stable matching.
- ▶ The boy in question was rejected by all the girls.
- ▶ Therefore, in any stable matching none of the girls are paired with him.
- ▶ Hence: the group of $n-m$ boys that aren't paired in BP-DAA is the group of boys not paired in any stable matching.

Don't be so picky...

Who is to blame for being unmatched?

- ▶ Who is to blame if a person remains unmatched?
- ▶ Maybe he is to blame. Maybe if changed his preferences he would find a mate.
- ▶ Perhaps an unemployed physicist would find a job, if he only put the municipal sanitation department, rather than Motorola, at the top of his list.
- ▶ No. A change of preferences will not change his position in any stable matching.

The unemployed, or the lonely, cannot help it...

- ▶ Denote by B the set of all boys who are not matched in any stable matching for a preference table T .
- ▶ Let T' be the preference table which differs from T only in the preferences of boys in B .
- ▶ In the GP-DAA according to T , no boy in B is visited by any girl.
- ▶ Therefore, in this process, preferences of boys in B do not come into play at all.
- ▶ Hence, the GP-DAA according to T' is exactly the same as the one according to T .
- ▶ That is, the boys in B are unmatched in the GP-DAA according to T' as well. Therefore they are unmatched in any stable matching.

Are there any other stable matchings?

- ▶ The question remains if there are other stable matchings besides the ones obtain by the boys proposing DAA and the girls proposing DAA.
- ▶ We've have already seen that if these two coincide, then there is a unique stable matching.
- ▶ But what if they don't coincide, as in the example we had before?
- ▶ In the example we had before, there is exactly one stable matching in addition to the girls propose DAA and Boys propose DAA.
- ▶ **Homework: Find this matching.**
- ▶ **Try to do it by yourself first and then check your answer in the next slide.**

The extra stable matching: solution

■ - girls propose

■ - boys propose

- In a stable matching:
- Adam is matched with either Ann or Carol, and Barry is matched with either Carol or Daisy.
- Suppose that Barry is matched with Daisy.
- Since Daisy prefers Dave to Barry, Dave must be matched with Ann.
- Therefore, Adam is matched with Carol. This is the outcome of BP-DAA.
- Suppose that Barry is matched with Carol. Adam is then matched with Ann.
- This can be completed to either the GP-DAA or by matching Claud-Betty, Dave-Daisy.
- Thus the only other stable matching is:
- Adam-Ann, Barry-Carol, Claud-Betty, Dave-Daisy.

	Ann	Betty	Carol	Daisy
Adam	3 1	4 1	2 3	1 4
Barry	2 4	1 4	4 1	3 3
Claud	2 2	1 3	3 2	4 1
Dave	1 3	4 2	3 4	2 2

Easy homework:

1. All boys prefer Cleo to all other girls.
 - ▶ Show that in any stable matching Cleo is matched to the same boy. Who is the lucky one?
2. Assume that there is the same number of boys and girls. Xantipah is last on every boy's preference list.
 - ▶ Show that in any stable matching Xantipha is matched to the same boy.
3. All boys have the same preference over girls. Show that there exists a unique stable matching. (Hint: just argue that the boys DAA and the girls DAA would give the same matching. Do an example first to convince yourself that this is the case.)

Placement of residents in hospitals, or polygamous matchings

- ▶ So far, we have assumed that every boy is permitted to date only one girl and that each girl is permitted to date only one man.
- ▶ We considered that case because it was the simplest to analyze.
- ▶ However, for many cases:
 - ▶ Placement of medical residents and hospitals,
 - ▶ Universities and students,
 - ▶ Corporations and employees;there are asymmetries between the two sides of the market:
 - ▶ Hospitals are interested in more than one resident, Schools in more than one student and employer in more than one employee,
 - ▶ while at the same time residents, students and employees chose one and only one hospital, university or employing corporation respectively.
- ▶ However, in each of these examples, the **places are limited**, each hospital, university or corporation have **quotas**, which is the maximum amount of residents, students or employees that they can possibly accept.
- ▶ We can extend the ideas that we have developed until now about stable matchings to this case.

Polygamous matchings: definitions

- ▶ Since we were interested in the problem of assigning residents to hospitals, let's keep working with that example.
- ▶ A polygamous matching problem is given by:
 - ▶ A finite set of hospitals and a finite set of students.
 - ▶ For each hospital, a quota that represents the maximal number of residents it can accept.
 - ▶ For each student, a ranking of hospitals.
 - ▶ For each hospital, a ranking of students.
- ▶ A matching of hospitals to residents is an assignment of each resident to a hospital, such that each resident is associated with at most one hospital, and for each hospital, the number of residents matched to it respects its quota.
- ▶ **We say that a hospital H and a resident r object to a matching if**
 - ▶ The resident r prefers hospital H to the one he/she is matched.
 - ▶ The hospital H is matched to fewer students than its quota limit, or if it meets its quota, then it prefers resident r to one of the residents that are currently matched to it.
- ▶ **A matching is stable if there is no pair of hospital and student who can object the matching.**

Stability of hospital-resident matchings

- ▶ Again, we have the same questions:
 - ▶ Firstly, if there is such a stable match, how to find it?
 - ▶ Secondly, do these matchings always exist?
- ▶ The answer is a resounding YES!
- ▶ Again, the **deferred acceptance algorithm (DAA)** described by Gale and Shapley (1962), altered a little bit, yields a stable matching.
- ▶ What is even more astounding, is that the initial National Resident Matching Program (NRMP) used since the 50's works in practice as the algorithm described by Gale and Shapley ten years later.
- ▶ In a paper from 1984, Alvin Roth studied the algorithm used by the NRMP and discovered that it was closely related to the Gale-Shapley algorithm. He then hypothesized that the fundamental reason for the success of the NRMP was that it produced stable matches.

Stability of hospital resident-matchings: hospitals propose DAA

- ▶ As described in the introduction of this lecture, the version of the NRMP that was used from 1953 to 1998 involves all students indicating the ranking of the hospitals they are interested in, hospitals ranking the students, all parties sending these preferences to a centralized clearinghouse which would then use this information to find a stable matching.
- ▶ To find such a stable matching, the centralized clearinghouse used an algorithm that is essentially a version of the **Deferred Acceptance Algorithm in which the hospitals are the proposers**.
- ▶ Again, by way of an example, we will see how this procedure works.
- ▶ Remember, each resident has a ranking of hospitals, each hospital has a ranking of all residents and also has a quota of the maximum number of residents it is willing to accept.

Hospitals propose DAA: example

- ▶ Suppose we have three hospitals, X, Y and Z and eleven students: a, b, c, d, e, f, g, h, i, j, k.
- ▶ The preferences of the hospitals for residents (from left to right), and its quota of residents (in round brackets) are:
 - ▶ X (5): j b a k g d c e f h i
 - ▶ Y(2): d b h a j f k e c i g
 - ▶ Z(4): f c j b h d e g i k a
- ▶ The preferences of medical resident applicants (from left to right) are:

a: ZYX	d: ZXY	g: XZY	j: XYZ
b: ZYX	e: ZXY	h: YXZ	k: XYZ
c: YZX	f: XZY	i: YXZ	

- ▶ So, remember, at each round of the DAA, an hospital will propose to the residents that are at the top of their rankings while respecting their quota, and then student's who have more than one hospital lined up will reject whichever hospital is not their favorite among those that have proposed.

Hospitals propose DAA: example

- ▶ X (5): j b a k g d c e f h i
- ▶ Y(2): d b h a j f k e c i g
- ▶ Z(4): f c j b h d e g i k a

a: ZYX d: ZXY g: XZY j: XYZ

b: ZYX e: ZXY h: YXZ k: XYZ

c: YZX f: XZY i: YXZ

		a	b	c	d	e	f	g	h	i	j	k
Stage 1		X	XYZ	Z	Y		Z	X			XZ	X
	Rejected		XY								Z	
Stage 2		X	Z	Z	YX		Z	X	YZ		X	X
	Rejected				Y				Z			
Stage 3		XY	Z	Z	XZ		Z	X	Y		X	X
	Rejected	X			X							
Stage 4		Y	Z	ZX	Z	X	Z	X	Y		X	X
	Rejected			X								
Stage 5		Y	Z	Z	ZX	X	Z	X	Y		X	X
	Rejected				Z							
Stage 6		Y	Z	Z	X	XZ	Z	X	Y		X	X
	Rejected					X						
Stage 7		Y	Z	Z	X	Z	ZX	X	Y		X	X
	Rejected						Z					
Stage 8		Y	Z	Z	X	Z	X	XZ	Y		X	X
	Rejected							Z				
Stage 9		Y	Z	Z	X	Z	X	X	Y	Z	X	X
	Rejected											

- ▶ Thus the matching is:
 - ▶ X: d f g j k
 - ▶ Y: a h
 - ▶ Z: b c e i
- ▶ This is a stable matching!

Residents propose DAA: example

- ▶ Same example, but now, residents are the proposers.
 - ▶ X(5): j b a k g d c e f h i
 - ▶ Y(2): d b h a j f k e c i g
 - ▶ Z(4): f c j b h d e g i k a
- ▶ The preferences of medical resident applicants (from left to right) are:

a: ZYX	d: ZXY	g: XZY	j: XYZ
b: ZYX	e: ZXY	h: YXZ	k: XYZ
c: YZX	f: XZY	i: YXZ	

- ▶ So, remember, at each round of the DAA, a resident will propose to his/her preferred hospital, hospitals will keep the residents that they prefer as long as they have enough quota, otherwise, they reject their least preferred residents until they meet their quota.
- ▶ Rejected residents, propose to their next preferred hospital and the process repeats.

Residents propose DAA: example

- ▶ X (5): j b a k g d c e f h i
- ▶ Y(2): d b h a j f k e c i g
- ▶ Z(4): f c j b h d e g i k a

a: ZYX	d: ZXY	g: XZY	j: XYZ
b: ZYX	e: ZXY	h: YXZ	k: XYZ
c: YZX	f: XZY	i: YXZ	

		X(5)	Y(2)	Z(4)
Stage 1		fgjk	chi	abde
	Rejected		i	
Stage 2		fgjki	ch	abde
	Rejected			

- ▶ Thus the matching is:
 - ▶ X: f g j k i
 - ▶ Y: c h
 - ▶ Z: a b d e
- ▶ This is a stable matching!
- ▶ However, notice that residents prefer this matching to the one obtained by the hospitals propose DAA.
- ▶ In this matching, all residents except *i* are in their highest ranked hospital

Hospitals propose DAA vs Residents propose DAA

- ▶ We have a similar result as before: the matching obtained by the residents propose DAA is preferred by residents to the one obtained by the hospitals propose DAA.
- ▶ We can't make the same statement about hospitals in this case: hospitals have preferences over individual students, but is not clear how a hospital would rank a **group of students**. Therefore, the question of whether hospitals are better or not when they propose is not easily answered.
- ▶ However, we can still say, as before, than when hospital propose, they get better individual candidates according to their preferences.
- ▶ We also have the result that if the hospitals propose DAA and the Residents propose DAA yield the **same result** then **there is a unique stable matching, obtained by either method**.
- ▶ In general, Hospital DAA and Residents DAA yield different matchings.
- ▶ Furthermore, there might be more stable matchings which are not obtained by either procedure.

Picky doctors and rural hospitals.

- ▶ Suppose that the rankings of residents towards hospitals are truncated, in the sense that some (picky) doctors prefer to remain unmatched (unemployed) than to go to a certain hospital.
- ▶ For example, hospitals in rural areas.
- ▶ Does the fate of the unmatched residents depend on the matching? Might an unmatched doctor be paired with a hospital in a different stable matching?
- ▶ The answer is:

No!

- ▶ The unmatched picky doctors are the same in every stable matching.
- ▶ Hospitals in rural areas cannot fill positions for residents, and some people argue that the matching mechanisms should be changed so that more doctors end up in rural hospitals. But the theorem says that it is impossible as long as stable matching is implemented.

The National Residents matching program (NRMP) now

- ▶ Despite its success, the NRMP still encountered problems.
- ▶ The number of female medical students has grown since the 50's, when the NRMP was first implemented.
- ▶ Since then, it has become increasingly common that dual-doctor couples looked for internships/residencies in the same region.
- ▶ The NRMP could not accommodate these requests, so by the late 80's, many doctors choose not to sue the mechanism: a sign that it was no longer delivering stable matches.
- ▶ THE NRMP, where hospitals proposed to students, was also criticized for systematically favoring hospitals over students.
- ▶ So in 1995, Alvin Roth was asked to help design an improved algorithm that would eliminate these problems.
- ▶ He formulated an algorithm, built on applicant proposals and designed to accommodate couples.
- ▶ The new algorithm, adopted by the NRMP in 1997, has worked well and now handles over 20,000 positions per year.

Similar ideas, different uses.

- ▶ The Gale-Shapley algorithm proved to be useful in other applications.
- ▶ Ideas based on the algorithm have been used to assign students to High-schools in New York City.
- ▶ In 2003, Al Roth and his colleagues helped redesign the admissions process, based on an applicant-proposing version of the DAS algorithm.
- ▶ It also proved successful.

Matching Theory saves lives: one-sided markets and organ donors

- ▶ The situations we described before involve a matching setting in which we could differentiate two sides of the market.
- ▶ Some important real world situations are one-sided.
- ▶ One extremely important example is the matching of kidneys and other human organs to patients in need of a transplant.
- ▶ Doing this process efficiently and quickly is of utmost importance.
- ▶ Not surprisingly, this problem was also studied by Shapley, albeit in a theoretical way.
- ▶ Shapley proposed an algorithm, called the **top-trading cycles** algorithm, that is very simple.
- ▶ It is based on an initial allocation of objects and subsequent swapping between participants.
- ▶ A challenge in the case of human organs is that some kidney-patient pairs may not be compatible and a complex multilateral swap may be needed.
- ▶ Using the ideas from Shapley, Al Roth also tackled this problem: using theory and experimental work and a careful design, Roth proposed a solution that nowadays is used in a number of U.S. states.
- ▶ For these reasons and many more, both Shapley and Roth were awarded the 2012 Nobel Prize in Economics.

Tackling new frontiers: market design

- ▶ As I discussed in the intro to this Module, the striking feature of these examples is that prices are not part of the process.
- ▶ However, the applications of matching theory have yielded the surprising conclusion that in many cases, prices might not be needed at all to have an actual market.
- ▶ As economists, we usually think that prices are required to make a market function in a decentralized way. But matching theory has shown to us that this is not necessarily the case.
- ▶ Furthermore, more theoretical work has come up with the insight that matching models that do include prices, work in much the same way and produce stable matches with broadly similar features.
- ▶ In fact, matching with prices is closely related to auctions, where objects are matched with buyers and prices are decisive.
- ▶ Research that relates matching algorithms with auctions has generated interesting theoretical results that are applicable in practice.
- ▶ An example of this is internet ad auctions: Search engine routinely auction out space for advertisers (whenever you use Google, you see the result of this at the top of the page in the sponsored ads).
- ▶ The insights from matching theory have permitted economists to design new auction formats.
- ▶ All of these gave rise to a new field of economics called “market design”, which aims to bring these insights to situations that are in need of a market or where a market doesn’t function properly.